

TAKAYAMA, Japan

WMO: 476170

Lat: 36.1561N

Lon: 137.2533E

Elev: 562

StdP: 94.75

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-7.1	-11.7	1.5	-2.4	-10.2	1.7	-2.6	5.4	1.6	5.1	1.5	1.1	90	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	33.0	23.0	31.6	22.5	30.1	22.0	24.1	29.7	23.6	28.7	23.0	27.8	3.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.9	18.8	25.6	22.2	18.1	25.2	21.7	17.6	24.9	76.1	29.9	73.6	28.8	71.5	28.0	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.3	4.7	-11.9	35.0	1.9	1.0	-13.3	35.7	-14.4	36.3	-15.5	36.9	-16.9	37.6		
(5)			-12.3	25.2	1.9	0.7	-13.6	25.7	-14.7	26.1	-15.8	26.5	-17.1	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-1.0	-0.2	3.9	10.1	16.1	20.4	24.5	25.5	21.0	14.6	7.8	2.1
	DBStd	9.65	2.52	2.99	3.31	3.87	2.80	2.33	2.24	2.11	3.13	3.22	3.26	3.02
	HDD10.0	1196	342	287	191	46	1	0	0	0	0	4	82	244
	HDD18.3	2855	600	520	448	247	78	8	0	0	12	123	316	503
	CDD10.0	1969	0	0	1	49	190	311	448	479	329	145	16	1
	CDD18.3	587	0	0	0	1	9	69	190	221	91	6	0	0
	CDH23.3	4076	0	0	0	17	155	407	1332	1656	485	24	0	0
	CDH26.7	1424	0	0	0	2	26	77	507	683	127	2	0	0
Wind	WSAvg	1.6	1.3	1.5	1.8	2.0	2.0	1.8	1.7	1.7	1.6	1.5	1.4	1.3
Precipitation	PrecAvg	1610	96	90	112	114	123	153	204	176	200	141	101	100
	PrecMax	2108	197	166	185	276	323	305	451	527	460	448	215	241
	PrecMin	1092	30	42	52	27	47	71	32	51	52	34	17	34
	PrecStd	255	38	35	39	54	60	54	117	103	97	81	44	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	12.0	18.5	25.7	29.4	30.6	34.5	34.9	31.8	25.7	19.6	13.4
		MCWB	4.2	5.4	9.2	13.5	16.4	19.7	23.3	23.6	22.1	18.4	13.3	9.4
	2%	DB	5.8	8.9	15.4	22.5	26.9	28.7	32.8	33.4	29.8	23.6	17.2	10.6
		MCWB	2.7	4.2	8.0	12.2	15.3	19.0	23.0	23.1	21.5	17.1	11.7	7.5
	5%	DB	4.1	6.5	12.9	20.2	25.0	27.4	31.4	32.1	28.1	21.8	15.1	8.6
		MCWB	1.7	3.0	7.0	11.5	15.1	18.6	22.5	22.8	20.8	15.9	11.2	6.3
	10%	DB	2.7	4.6	10.4	17.8	23.1	25.8	29.7	30.7	26.3	19.9	13.3	6.7
		MCWB	0.9	2.0	5.8	10.9	14.3	18.4	22.2	22.6	20.2	15.2	10.7	4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	7.0	11.7	17.2	19.7	22.7	25.0	25.0	23.8	20.5	15.3	10.8
		MCDB	7.1	10.2	14.4	18.7	22.7	26.1	30.9	31.2	28.4	23.3	17.2	11.8
	2%	WB	3.1	5.0	9.5	14.6	18.2	21.8	24.1	24.2	22.7	18.8	13.5	8.3
		MCDB	4.9	7.6	13.4	18.8	22.3	25.0	29.8	29.9	26.9	21.5	15.1	10.0
	5%	WB	2.0	3.5	7.9	13.3	17.0	20.9	23.5	23.7	22.0	17.4	12.1	6.4
		MCDB	3.6	5.7	11.3	17.6	21.7	23.9	28.7	29.0	25.8	20.2	14.3	8.3
	10%	WB	1.2	2.3	6.3	12.0	15.9	19.9	22.9	23.3	21.3	16.1	10.9	4.8
		MCDB	2.4	4.2	9.7	16.3	20.6	23.4	27.7	28.3	24.7	18.9	13.0	6.4
Mean Daily Temperature Range	5% DB	MDBR	7.3	9.0	10.7	12.8	12.7	10.4	9.4	10.1	9.4	10.1	9.3	7.2
		MCDBR	9.2	12.2	15.6	17.2	16.6	13.1	12.2	12.3	11.4	11.9	12.2	10.4
		MCWBR	6.3	7.8	8.6	7.8	6.8	5.0	4.0	3.9	4.2	5.6	7.2	7.0
	5% WB	MCDBR	8.2	10.6	12.8	12.3	12.2	8.9	10.7	10.5	9.4	9.2	9.9	9.5
		MCWBR	5.9	7.3	8.4	6.7	6.2	4.2	3.8	3.7	4.0	5.0	6.6	6.9
Clear-Sky Solar Irradiance	taub	0.373	0.410	0.481	0.514	0.520	0.547	0.529	0.522	0.462	0.420	0.389	0.371	
	taud	1.981	1.901	1.833	1.871	1.919	1.922	2.044	2.063	2.187	2.214	2.207	2.100	
	Ebn at Noon	780	800	779	779	781	758	769	765	793	790	770	759	
	Edh at Noon	136	166	195	198	192	191	168	162	136	121	109	114	
All-Sky Solar Radiation	RadAvg	1.94	2.72	3.69	4.60	5.24	4.83	4.72	4.85	3.87	3.17	2.26	1.69	
	RadStd	0.18	0.19	0.28	0.40	0.58	0.37	0.70	0.50	0.32	0.29	0.19	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page